



Hikvision Fingerprint to Odoo Attendance

Integration Guide — Option B (FastAPI Bridge)

Odoo: <https://shamsieh-testing-33874628.dev.odoo.com>

Webhook: <http://192.168.100.4:8080/hikvision/attendance>

Document date: July 07, 2026



1. Overview

This integration connects a Hikvision fingerprint terminal on the local network to Odoo Attendances on Odoo.sh. A FastAPI bridge runs on a Windows PC, receives device events, and creates hr.attendance check-ins through XML-RPC. No Fingerprint Device configuration is required inside Odoo.

2. Architecture

Component	Role
Hikvision device (192.168.100.85)	Captures fingerprint; pushes HTTP events
FastAPI bridge (192.168.100.4:8080)	Parses events, idempotency, retries
Odoo.sh	Stores employees and attendance records
SQLite (local)	Processed events + retry queue

Flow: Device POST → /hikvision/attendance → find employee by barcode → create check-in.

3. Configuration

3.1 Hikvision HTTP Listening

Field	Value
Event Alarm IP/Domain Name	192.168.100.4
Port	8080
URL	/hikvision/attendance
Protocol	HTTP

3.2 Odoo Employee Mapping

- RFID/Badge Number = Hikvision employeeNoString (example: 5)
- Biometric Device User ID = same value (recommended)

3.3 Bridge Environment (.env)

Variable	Example
ODOO_URL	https://shamsieh-testing-33874628.dev.odoo.com
ODOO_DB	shamsieh-testing-33874628
ODOO_BOT_USER	admin
ODOO_API_KEY	API key from Odoo Preferences
LISTEN_PORT	8080



4. HTTP API & Status Codes

The bridge returns meaningful HTTP status codes so operators and integrators can distinguish success, wrong fingerprint, missing employee, and system events.

4.1 Endpoints

Method	URL	Purpose
GET	/health	Service health check
GET	/odoo/ping	Test Odoo.sh XML-RPC connection
GET	/hikvision/attendance	Webhook reachability check
POST	/hikvision/attendance	Receive Hikvision fingerprint events

4.2 POST /hikvision/attendance — Status Codes

HTTP	reason / result	When it happens
201	created	Fingerprint success; new hr.attendance check-in created
200	duplicate-attendance	Employee already checked in today
200	duplicate-event	Same device event already processed (idempotency)
404	employee-not-found	employeeNoString not found in Odoo barcode
422	fingerprint-failed	Wrong finger / failed scan (subEventType 39)
422	missing-employee	Event has no employee number
422	not-fingerprint-event	Not a fingerprint authentication event
204	system-event	Door/system event (subEventType 21–24); ignored
400	invalid-payload	Malformed or unparseable request body
503	odoo-unavailable	Odoo.sh unreachable; event queued for retry

4.3 Hikvision Event Codes

subEventType	Meaning	Bridge action
38	Fingerprint verification success	Process attendance if employee mapped
39	Fingerprint verification failed	HTTP 422 fingerprint-failed
21	Door locked	HTTP 204 system-event (ignored)
22	Door unlocked	HTTP 204 system-event (ignored)

5. Postman Test Bodies



5.1 Success (expect 201)

POST http://192.168.100.4:8080/hikvision/attendance

Content-Type: application/json

```
{"subEventType":38,"employeeNoString":"5","currentVerifyMode":"fp","dateTime":"2026-07-07T13:40:51+03:00","serialNo":9999,"status":"success"}
```

5.2 Wrong fingerprint (expect 422)

POST http://192.168.100.4:8080/hikvision/attendance

Content-Type: application/json

```
{"subEventType":39,"currentVerifyMode":"faceOrFpOrCardOrPw","status":"failed","serialNo":9998}
```

6. Service Features

- Parses Hikvision multipart/form-data JSON (event_log field).
- Uses webhook receive time when device omits dateTime.
- SQLite idempotency on (device_serial, event_id).
- Retries failed Odoo calls every 30 seconds.
- Auto-start via Windows scheduled task HikvisionAttendanceBridge.

7. Start & Verify

- Manual: .\run.ps1 in scripts/hikvision_attendance_service
- Health: GET http://192.168.100.4:8080/health
- Odoo: GET http://192.168.100.4:8080/odoo/ping
- Fingerprint on device → check Attendances in Odoo.sh

— End of Document —